Eldad Tolla

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Education

Columbia University, School of Engineering and Applied Science New York, NY
B.S. in Computer Science and Mathematics *Expected May 2027*

- **Honors:** Dean's List (All Semesters), NSF SUPER Scholar, CS + Math Most Outstanding Student
- **Coursework:** Data Structures, Algorithms, Machine Learning, Reinforcement Learning, Linear Algebra, Probability, HPC

Technical Skills

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, R
Frameworks & Libraries: React, Next.js, Node.js, Flask, PyTorch, TensorFlow, LangChain
Infrastructure & Tools: AWS, Supabase, Neo4j, SQLite, Git, Docker, REST APIs, CI/CD, Agile
Specializations: Full-Stack Development, LLM Orchestration, RAG Pipelines, HPC (TACC)

Experience

AI Agent Builder Intern Aug 2025 – Present
NeuralSeek *Remote*

- Built and deployed **10 production AI agents** for 2 enterprise clients, automating policy, compliance, and document-search workflows using tool-augmented LLM orchestration
- Integrated OpenAI LLMs with vector databases (**10K+ documents**), building a RAG pipeline that improved search relevance by **30%** and reduced hallucinations by **40%**
- Optimized agent pipelines through prompt routing and retrieval tuning, cutting response latency by **25%** and improving answer accuracy by **20%** across enterprise workflows

Research Intern — NSF Cyberinfrastructure REU May 2025 – Aug 2025
Texas Advanced Computing Center (TACC), UT Austin *Austin, TX*

- Engineered warm-start Q-learning agents with search-based planning priors, achieving **38–62% faster convergence** and **67–133% higher reward** vs. cold-start baselines
- Designed and ran **15 independent experiments** across 1,000 episodes on **90 GPU HPC nodes**, benchmarking initialization strategies and identifying DFS as more stable and ClosestDot as faster-converging for warm-start selection
- Authored research paper on hybrid planning + RL and presented findings via poster and oral talk at TACC REU symposium to 50+ researchers

IT Help Desk Assistant Aug 2024 – Aug 2025
Randolph College *Lynchburg, VA*

- Resolved **20+ weekly support tickets** for 300+ users while maintaining **100% SLA compliance** (24-hour resolution)
- Built a digital inventory system tracking **200+ devices**, reducing equipment retrieval time by **40%**

Projects

Lisan — Language Learning Platform | *Next.js, React, Supabase, TypeScript* 2025 – Present

- Architected and built a **full-stack language learning platform** for underrepresented languages, starting with Amharic, using Next.js frontend and Supabase backend (auth, database, storage)
- Designed scalable data models (letters → words → phrases → exercises) and an **audio-first learning pipeline** with sub-second pronunciation playback latency
- Implemented **WCAG-compliant** accessibility-first UX with high-contrast UI, touch-friendly controls, and multi-modal support for neurodiverse learners

Sentinel — Multi-Agent Investigative AI | *Python, Neo4j, Dedalus ADK, ElevenLabs* DivHacks 2026

- Led AI/ML development of a **4-agent pipeline** (Pattern Finder → Timeline Builder → Evidence Ranker → Report Compiler) that analyzes government finance data from a Neo4j graph database
- Engineered LLM prompt chains producing **citation-backed investigative reports** with auto-generated NPR-style audio narration via ElevenLabs; implemented caching fallback achieving **100% demo reliability**

CommunityWatch — Civic Tech Platform | *Next.js, Supabase, Leaflet, TypeScript* 2025

- Built a full-stack civic reporting platform with Supabase auth, location-tagged issue reporting, and an interactive Leaflet map overlaying HOLC redlining data to surface service delivery disparities
- Developed auto-advocate tool that generates 311-ready complaint text with one-click submission; shipped full platform end-to-end in **48 hours**

Leadership

Judiciary Chair | Randolph College Aug 2024 – Present

- Mediated **30+ student conduct cases** per semester; implemented restorative justice practices reducing repeat offenses by **40%**